

Roll No.:

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech

Semester: III

Name of the Course: Machine Learning

Course Code: TCS509

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1		(20Marks)																											
(a)	Explain the concept of Machine Learning. Discuss the types of Machine Learning with examples. How do Supervised and Unsupervised Learning differ in their approach and applications?		CO1																										
(b)	What do you mean by gain and entropy? How it is used to build the decision tree.		CO1																										
(c)	Explain the logistic regression model. Derive the loss function using cross-entropy.		CO2																										
Q2																													
(a)	Describe bagging and boosting ensemble learning with example. What is Random Forest? Random forest is an example of which ensemble learning.		CO2																										
(b)	Describe the Elbow Method and Silhouette Scores for determining the optimal number of clusters in K-means. Illustrate their application with a suitable example.		CO3																										
(c)	Use the k-means algorithm and Euclidean distance to cluster the following example into 2 clusters:		CO3																										
	<table><tr><th>Height</th><th>Weight</th></tr><tr><td>185</td><td>72</td></tr><tr><td>170</td><td>56</td></tr><tr><td>168</td><td>60</td></tr><tr><td>179</td><td>68</td></tr><tr><td>182</td><td>72</td></tr><tr><td>188</td><td>77</td></tr><tr><td>180</td><td>71</td></tr><tr><td>180</td><td>70</td></tr><tr><td>183</td><td>84</td></tr><tr><td>180</td><td>88</td></tr><tr><td>180</td><td>67</td></tr><tr><td>177</td><td>76</td></tr></table>		Height	Weight	185	72	170	56	168	60	179	68	182	72	188	77	180	71	180	70	183	84	180	88	180	67	177	76	
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Q3																													
(a)	Explain Naïve Bayes Algorithm with examples. Consider the data set shown in table below.		CO3																										
	<table><tr><th>A</th><th>B</th><th>Class</th></tr><tr><td>1</td><td>0</td><td>Positive</td></tr><tr><td>0</td><td>0</td><td>Negative</td></tr><tr><td>0</td><td>1</td><td>Negative</td></tr></table>		A	B	Class	1	0	Positive	0	0	Negative	0	1	Negative	CO4														
	A	B	Class																										
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		CO4																											

		0	1	Negative		
		0	0	Positive		
		1	0	Positive		
	Estimate all the parameters for training Naïve Bayes classifier and classify the given test sample (A=1, B=1) using estimated parameters.					
(b)	Provide an overview of reinforcement learning. How does it differ from supervised and unsupervised learning paradigms?					
(c)	Describe Agglomerative and Divisive Hierarchical Clustering. Compute Agglomerative Hierarchical Clustering in a given distance matrix.					
		A	B	C	D	E
	A	0				
	B	0.71	0			
	C	5.66	4.95	0		
	D	3.61	2.92	2.24	0	
	E	4.24	3.54	1.41	1.0	0
Q4						
(a)	What are autoencoders? Describe their structure and provide an example of their real-world application.					CO4
(b)	Explain PCA and its process with their applications? Consider the two-dimensional pattern: (1,1), (1,2), (4,4), (5,5). Use the PCA to transform the pattern (4,4).					CO4
(c)	Explore activation functions in neural networks. Provide examples of one non-linear activation function and one linear activation function, and discuss their significance.					CO5
Q5						
(a)	Draw and explain the architecture of Convolutional Neural Network.					CO5
(b)	Discuss the concept of backpropagation in training neural networks. Provide an example of how weights are updated during the process.					CO6
(c)	Write the short notes on the following Deep learning tools: (i)TensorFlow (ii) Word2Vec (iii) AlexNet					CO6